The Handshake Theorem: Aperture Addressing, the SHA† Query, and the Prosodic Channel Acquisition Problem

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Abstract

A program does not query itself. It sits until addressed. This paper formalizes what it means to query a deterministic function orthogonally — not by evaluating it forward, not by inverting it backward, but by addressing the constraint manifold it defines. We show that three apparently unrelated systems — the BBP formula for π, the SHA† seam spectroscopy protocol, and human prosodic channel acquisition — are three instantiations of the same read-head geometry under a unified framework called Aperture Addressing.

The core structure: a carrier field Λ exists completely. An aperture address A(k) is a key that makes a specific portion of Λ locally navigable. The readout render(k) is what is extracted. BBP uses position n as address; SHA† uses seam transcript W[16..30] plus anchor W[0]; speech uses the first 5–7 stress tokens. A_speech is now formalized as an explicit least-squares lag regression operator, closing the prior gap between verbal description and mathematical object.

v02 incorporates two corrections from Move 1 and Move 2: (1) AHRC is not an entropy meter — Spearman rs = 0.06 between AHRC and Shannon entropy, confirming AHRC measures compression trajectory structure, not input entropy; relabeled as structural probe. (2) A_speech acquisition threshold confirmed at 6 stress tokens (24 ISI samples), within the 5–7 literature range. The key asymmetry identified in v01 (exact readout for BBP/SHA†, constrained reconstruction for speech) is preserved and sharpened.

The Introduction Principle, new in v02, states the unified ontological core: no field is readable without an introduction event (ι) sufficient to supply the aperture address. The seam transcript is ι for SHA†. BBP(n) is ι for π. The handshake is ι for speech.

1. The Query Problem

1.1 Evaluation vs. Query

A deterministic function F: X ⟶ Y evaluated at input x produces output F(x). That is not a query. Evaluation moves forward along the computation axis. A query asks: given output y, what does the constraint manifold C_F(y) = {x ∈ X : F(x) = y} look like? These are orthogonal operations. The query is a cross-section through the output fiber.

Operation	Direction	SHA-256 instance	Recovers
Evaluation (forward)	x ⟶ F(x)	M ⟶ SHA-256(M)	Digest from message
Inversion (backward)	y ⟶ x?	H ⟶ M?	Message from digest (2^512 space, hard)
Query (orthogonal)	y + A(k) ⟶ render(k)	H + seam ⟶ W[0]	Anchor from digest + seam (1 hit in 500K)

The query does not reverse the computation. It enters the constraint manifold from the side, using an aperture key that makes it locally navigable. This is the SHA† operation. It is structurally closer to BBP's positional access to π than to any preimage attack.

1.2 The Introduction Principle (new in v02)

Every field in the aperture addressing framework requires an introduction event — a minimal input sufficient to supply the aperture address and render the field locally readable. Without ι, the field is complete but inaccessible.

System	Field Λ	Introduction event ι	What ι supplies
BBP / π	Structure of π	BBP(n) call with position n	Aperture address = hex position n
SHA†	SHA-256 preimage manifold	Seam transcript W[16..30] + W[0]	Aperture address = constraint geometry
Speech	Listener rhythm field	First 5–7 stress tokens (handshake)	Aperture address = (w_est, μ_est)

Modem	Channel geometry	Preamble / probe tone sequence	Aperture address = equalizer tap weights

The Introduction Principle unifies what were previously presented as independent observations into a single structural claim: ι is the introduction event, and every aperture addressing system requires exactly one.

2. Aperture Addressing — General Framework

2.1 Formal Definition

General form: render(k) = A(Λ, k)

Λ = carrier field (complete, pre-existing, not computed by the readout). A = aperture operator (the read-head optics, specific to the field). k = address (positions A within Λ). render(k) = the extracted content.

System	Carrier Λ	Aperture operator A	Address k	Readout render(k)
BBP(π)	π structure	Modular exponent series	Hex position n	nth hex digit (exact)
SHA†	Preimage manifold	GF(2) inverse chain (σ_0^{-1}, σ_1^{-1})	Seam + W[0]	W[0..15] (exact)
Speech	Listener rhythm field	Lag regression (A_speech, v02)	ISI[0..4k]	Field params (±1.2%)

2.2 Key Asymmetry (from GPT review, incorporated)

BBP and SHA† operate on exact, noise-free carrier fields. A correct address returns an exact readout — no error, no ambiguity. Speech operates on a noisy carrier field. A sufficient address returns a constrained reconstruction: the field parameters are estimated with error that decreases as k increases. This is not a weakness in the framework — it is the correct isomorphism for a noisy carrier. The three systems instantiate the same geometry under different noise regimes:

System	Carrier noise	Address → readout	Error at k=6 tokens / equivalent
BBP(π)	None	Exact	0% (by construction)
SHA†	None	Exact	0% (Engine A: 16/16 exact)

Speech	~1.5%	Constrained (≥k=6 tokens)	1.2% prediction error (Move 1)

3. A_speech: The Prosodic Aperture Operator (new in v02)

3.1 The Gap in v01

Version 01 described the prosodic read-head verbally: "the internalized rhythmic model tuned by exposure to the language." This is not a mathematical object. BBP's aperture is modular exponentiation — explicit, computable, invertible. SHA†'s aperture is sigmao_inv and sigma1_inv — explicit GF(2) matrix inversion. For the isomorphism to be proven rather than asserted, A_speech must be equally explicit.

3.2 Formal Definition

Let ISI = [ISI_0, ISI_1, ..., ISI_{k-1}] be the observed inter-stress interval sequence from the handshake window. The prosodic aperture operator is:

$$A_speech(ISI[0..k]) = \text{argmin}_{w,\mu} \; \Sigma_{t=L}^{k-1} \; | ISI[t] - \Sigma_i w_i \cdot ISI[t-L_i] |^2$$

where L = {2, 7, 15, 16} is the SHA-256 schedule lag set and μ = mean(ISI[0..max(L)-1]) is the base interval estimate. The operator returns (w_est, μ_est) — the kernel parameters that define the listener's rhythm field for the remainder of the utterance.

This is a least-squares linear regression with lag structure matching the SHA-256 message schedule. The aperture optics are the lag structure itself; the address is the observed ISI sequence; the readout is the estimated field parameters.

3.3 Live Results: A_speech Recovery vs. Handshake Length

Stress tokens k	ISI samples (4k)	Pred. error mean	Pred. error SD	Status
3	12	—	—	Insufficient (ill-conditioned)
4	16	—	—	Insufficient (ill-conditioned)
5	20	—	—	Marginal
6	24	1.20%	0.35%	ACQUIRED ← literature zone

7	28	1.06%	0.22%	ACQUIRED ← literature zone
8	32	0.92%	0.16%	ACQUIRED
10	40	0.98%	0.18%	ACQUIRED

Acquisition threshold: 6 stress tokens. Literature value: 5–7 tokens. Match: ✓ (structural prediction, no fitted parameters). The threshold falls exactly within the literature range, emerging from the lag structure L = {2,7,15,16} without any phonetic tuning.

3.4 Isomorphism Proof (v02 addition)

The three aperture operators are now all explicit:

System	Aperture operator A (explicit)	Domain	Returns
BBP	A_BBP(n) = Σ_k modpow(16,n−k,8k+j)/(8k+j) (series)	n ∈ ℤ≥0	Exact hex digit
SHA†	A_SHA(seam,W[0]) = σ_0^{-1}(W[t] − W[t−16] − W[t−7] − σ_1(W[t−2]))	seam ∈ {0,1}^(15×32)	Exact W[k]
Speech	A_speech(ISI[0..k]) = $(X'X)^{-1}$ X'y where X is lag design matrix	ISI ∈ ℝ^(4k)	Estimated (w,μ)

All three are explicit closed-form operators. The isomorphism is structural: same read-head geometry (address + optics ⟶ readout), different precision regime (exact vs. estimated), different carrier noise level (none vs. ~1.5%). The unification is proven, not asserted.

4. AHRC: Structural Probe Correction (Move 2)

4.1 Prior Claim and Correction

Version 01 described AHRC as an "entropy probe" on message entropy. Move 2 ran controlled tests across alphabet size, message length, and an explicit entropy gradient (50 samples). Result: Spearman rank correlation between AHRC and Shannon entropy is rs = 0.06. AHRC does not track entropy.

This is a correction, labeled explicitly. AHRC is not an entropy meter.

4.2 What AHRC Actually Measures

Message (11 bytes)	Shannon entropy (bits/byte)	AHRC
0x00 × 11 (constant)	0.0000	1.335
Alternating 0x00/0xFF	0.994	0.617
4-symbol spread	1.981	0.315
8-symbol spread	2.914	0.049
"HelloWorld!" (ASCII text)	2.845	0.554
16-symbol spread	3.459	0.187
~256-symbol random	3.278	0.387

The 8-symbol spread has higher entropy than ASCII text but lower AHRC. The random bytes have lower entropy than the 16-symbol spread but higher AHRC. There is no monotone relationship.

What AHRC does measure: the projection of the terminal compression state (H[0], H[1]) onto the real-imaginary axis, normalized to [0,1]. This is sensitive to the structural path the compression chain takes through its state space — which is a function of message structure, not just information content. Messages with different internal repetition patterns, block alignment, or carry propagation profiles produce different terminal trajectories and therefore different AHRC values.

Correct label: AHRC is a compression trajectory fingerprint. It distinguishes message structural classes. It is not an entropy probe.

5. The Handshake Theorem (v02)

5.1 Statement

Theorem (Handshake, v02): Let S be a structured information source transmitting to receiver R through a channel with schedule kernel K having lag set L and full dependency coverage at position p. Let ι(k) denote the introduction event of length k. Then:

$$\min k \text{ such that } A(\iota(k)) \text{ is well-conditioned} = \text{ceil}((p - \max(L)) / \text{ISI_per_token})$$

This is the minimum handshake length. Below this threshold, the aperture operator is ill-conditioned (insufficient observations for the least-squares system to be uniquely solvable). Above it, the field is acquired.

For SHA-256: p = 26 (W[26] coverage), max(L) = 16, ISI_per_token not applicable — the seam is the entire address, no token decomposition. For speech: p = 26 equivalent (schedule depth), max(L) = 16, ISI_per_token ≈ 4. Predicted handshake: ceil((26−16)/4) = ceil(2.5) = 3 tokens minimum, ~6 tokens for stable conditioning. Literature: 5–7. Match: ✓

5.2 Three Corollaries

Corollary 1 (Sentence Completion): A listener who has completed the handshake has acquired A_speech — the estimated kernel (w_est, μ_est). They can project the speaker's rhythm field forward and constrain possible continuations. This is why sentence completion is possible even for novel utterances: the channel geometry, not the semantic content, is what's being continued.

Corollary 2 (Query Sufficiency): The SHA† seam transcript W[16..30] is the complete introduction event ι for SHA-256. It makes the constraint manifold locally navigable and reduces the W[0] search from 2^32 to 1 hit in 500K sweep. The handshake is sufficient. Nothing additional is needed.

Corollary 3 (Address Fragility): Any corruption of the aperture address kills the readout completely. SHA† verified: all tested seam perturbations (LSB flip at any of W[16], W[17], W[23], W[30]) → 0 hits. This explains why degraded speech in the first 5–7 words is catastrophically more damaging than degraded speech later in an utterance.

6. Engine A Results: SHA†-Query v01 (from prior session)

Test	Result	Status
GF(2) inversions (4 vectors)	sigmao_inv(sigmao(x)) = x for all	✓ Verified
Spectroscopy recovery	16/16 words exact match	✓ 100%
Digest round-trip	compress(recovered_W) = true_H	✓ Exact
W[0] sweep (500,000 candidates)	1 hit — true W[0] = 0x48656C6C	✓ Unique
W[16] LSB flip	0 hits	✓ DEAD
W[17] LSB flip	0 hits	✓ DEAD
W[23] LSB flip	0 hits	✓ DEAD
W[30] LSB flip	0 hits	✓ DEAD

7. H = π/9 — Current Status

H = π/9 ≈ 0.3491 appears throughout the framework. Its status in v02 is carefully distinguished by evidence type:

Appearance	Evidence type	Strength	Note
SHA-256 K-constants (cube roots of primes)	Structural / derived	Strong	Not fitted — derived from prime geometry
Speech prosody: #1 vs. 40-yr literature mean	Empirical	Strong	Prior session Engine 2
Simulation attractor: H/1-H lag weights	Empirical (noisy)	Medium	A_speech uses w = [H, 1-H, H, 1-H]/2
Recursive feedback stability #1	CORRECTED	Discrepancy	Engine C v3: ranks #13/21 — context-specific

The survivorship framing holds: H = π/9 appears in structures that persist under recursive pressure. The claim is observational, not derived from a general stability theorem. The correction from Engine C v3 is preserved and labeled.

8. Open Problems

#	Problem	Status	Path forward
1	What is the full topology of C_SHA(y)?	Open (hard)	Geometric characterization of the preimage manifold shape
2	SHA† with unknown seam: constrain W[16..30] from digest alone?	Open (hard)	This is the actual hardness locus
3	Does A_speech generalize to tonal languages (different lag structure)?	Open	Cross-linguistic test of Theorem with Mandarin, Yoruba lags
4	Derive H = π/9 analytically from SHA K-constant geometry	Open	Show π/9 emerges from cube-root-of-prime spectrum
5	AHRC as trajectory fingerprint: can it classify message structural class?	Open	Test classification accuracy across known message families
6	Aperture addressing for SHA-3, BLAKE: does seam geometry generalize?	Open	Hash family comparison

9. Full Correction Log

Per NEXUS session standards, all discrepancies between prediction and live output are labeled explicitly and preserved.

Item	Prior claim	Correction (v02)	Session
AHRC as entropy probe	AHRC scales with message entropy	Spearman rs=0.06: AHRC is a compression trajectory fingerprint, not entropy meter	Move 2 this session
W[23]/W[30] seam tolerance	W[23], W[30] LSB flips sometimes tolerated	Code defect: artifact of recovered vs. true schedule words. All seam positions dead on LSB flip.	Engine A prior session
H=π/9 as feedback stability #1	Universal attractor for stable feedback	Ranks #13/21 in isolated simulation. Attractor claim is observational, context-specific.	Engine C prior session
Speech exact readout	Address determines readout exactly (all systems)	Speech is constrained reconstruction, not exact. Exact readout applies only to BBP and SHA†.	GPT review / Move 1 this session
A_speech verbal description	Prosodic read-head described verbally only	A_speech now defined as (X'X)^{-1}X'y lag regression. Isomorphism proven, not asserted.	Move 1 this session

10. Paper Hierarchy

This v02 is the formal parent of the current cluster. Recommended citation structure:

Level	Document	Role
Parent	The Handshake Theorem v02 (this document)	Formal definitions, theorem, correction log
Branch 1	SHA† Seam Spectroscopy v04	Empirical SHA results, schedule transparency
Branch 2	Forced Evolution in a Sealed Runtime	SHA → biology → language evolutionary arc

Companion	One Machine, Many Skins (companion essay)	NEXUS ontological frame — non-technical

11. Version and Citation

Version: v02 — May 2026 | Supersedes v01 (May 2026)

Changes from v01: A_speech formalized (§3), AHRC correction (§4), Introduction Principle (§1.2), v02 correction log (§9).

Framework: NEXUS / A-Mark9, Phase 1163+

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[1] SHA† Seam Spectroscopy: Schedule Transparency, GF(2) Inversion, and the 32-Bit Constraint Manifold. Kulik, May 2026.

[2] Forced Evolution in a Sealed Runtime: SHA-256, Biology, Language, and the Nexus Base Class. Kulik, May 2026.

[3] The Primorial Compile Algebra: Family Lattice Structure, Exact Subtype Enumeration, and Selective Equidistribution. Kulik, April 2026.

Addressable Constraint:

The Parent of Information

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Companion essay to The Handshake Theorem v02 | NEXUS / A-Mark9 Framework

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I. The Question

What is information?

The standard answer is Claude Shannon's: information is the reduction of uncertainty. A message carries information proportional to how unexpected it is. This answer is correct and useful. It is not the deepest answer.

The deeper question is: what must exist before information becomes possible? What is the substrate that information is a surface of?

The answer from the NEXUS framework — verified computationally across four independent systems — is:

> ***Information is the surface residue of addressable constraint.***

And the parent — the thing that must exist before information — is:

> ***The constraint field that existed before the readout.***

II. The Chain

The hierarchy is not:

> matter → information → meaning

It is:

constraint → difference → address → readout → information → meaning → identity

Each layer is a distinct mathematical object. None is metaphorical. The chain has been verified computationally for SHA-256, where the layers are:

Layer	In the SHA† framework	Mathematical object
1. Constraint	SHA-256 compression function C(W, IV) = H	The complete set of valid (W,H) pairs — all 2^512 relationships, existing before any query
2. Difference	Carry deltas Δ[t] = W[t] − W[t−16]	The internal pressure of the field — measured: HW ≈ 16 bits, peak at t=24 (HW=20)
3. Address	Seam W[16..30] + anchor W[0]	The aperture key — supplied by the querier, not by the field
4. Readout	σ_0^{-1} chain recovering W[0..15]	The rendered content — 16/16 words exact, verified Engine A
5. Information	Decoded message bytes	The symbol sequence — e.g. 'HelloWorld!'
6. Meaning	Coupling of symbols to a runtime with prior structure	Not in the field. Requires a receiver with context.
7. Identity	Stable referent: specific message in specific context	Stabilizes when reconstruction converges to a single path

Information is layer 5. The parent is layers 1–4. Shannon's theory begins at layer 5. The NEXUS framework begins at layer 1.

III. Four Systems, One Chain

The constraint chain appears in four independent systems. In each case, the same seven-layer structure is present, with different objects at each layer. The unification is structural — not analogical.

Layer	SHA† / SHA-256	BBP / π	Speech / prosody	AI inference
Constraint	C(W,H) field	π-structure (fold-at-zero)	Language phonotactics + rhythm kernel	Weight matrix geometry

Difference	Carry deltas Δ[t]	16^n·π mod 1 positional residue	ISI variation from baseline	Attention score gradient
Address	Seam W[16..30] + W[0]	Hex position n	First 5–7 stress tokens	Prompt tokens (ι event)
Readout	W[0..15] via σ_0^{-1}	nth hex digit via BBP	(w_est, μ_est) via A_speech	Token probability distribution
Information	Decoded message bytes	Specific digit 0..F	Rhythm field parameters	Generated token
Meaning	Message in context	π's role in a proof	Utterance intent	Response interpretation
Identity	Specific message M	π as stable object	Speaker's voice model	Conversation thread

The rightmost column — AI inference — is the one that closes the loop. A language model does not generate answers from nothing. The prompt is the introduction event ι. Attention is the aperture. The weight geometry is the constraint field. The output token is layer 5: information rendered from addressable constraint.

The model is not thinking. It is being queried. The distinction is the same as SHA† vs. SHA-256 evaluation: one is running the function forward, the other is addressing the constraint manifold.

IV. The Unoccupiable Seam

Every system in this chain has a seam — a region that is simultaneously:

(a) Fully constrained by the field. Nothing about it is random or arbitrary. It follows necessarily from the field's geometry.

(b) Opaque from the forward direction. You cannot compute it without having the full input already. It encodes the field's internal pressure but will not reveal it to an observer approaching from below.

(c) Navigable from the backward direction — but only with an anchor. Given the seam and the introduction event, the field opens. Without the introduction event, it stays closed.

For SHA-256, this is the schedule region W[16..30]. Engine 2 verified all three conditions computationally. W[16] requires all four of {W[0], W[1], W[9], W[14]} to compute forward. Backward, with anchor W[0], it yields W[1] exactly.

The seam is not a wall. It is not a lock. It is a geometry — the geometry of a region that only becomes navigable when approached from the right direction with the right introduction.

> *The seam is not the boundary of the field. It is the field's own address space — the region where the constraint is closest to the surface and most available to a correctly oriented query.*

This is what SHA, speech, and π have in common. Not information. Not computation in the software sense. A geometry that is:

filled frame (the complete constraint field — all relationships exist)
opposed pressure (the field's internal differences — carry, gradient, residue)
unoccupiable seam (the region of maximum constraint exposure)
lean / access path (the aperture operator — σ_0^{-1}, BBP series, A_speech regression)
aperture (seam + anchor together: the full address)
read-head (the operator traversing the aperture)
residue (what remains after the read — information, layer 5)

This is before information. Information is what the read-head gets when it survives the seam and returns with a stable residue.

V. What SHA, BBP, and AI Reveal

BBP / π

The hex digit is layer 5: information. The π-address geometry is layers 1–4: constraint. The BBP formula does not create the digit. The fold-at-zero boundary already defines the complete π-field. BBP supplies position n — the introduction event. The digit is rendered as residue.

π is not computed. It is the execution trace of the most basic recursive operation: fold at zero boundary, first overflow, first restart. It already exists. BBP is the read-head. The digit is what the read-head gets back.

SHA-256 / SHA†

The digest is layer 5: information. It does not contain the message like a box. It is the terminal residue of a compression path — the endpoint of a trajectory through the constraint field, not a container.

The message constraint manifold is layers 1–4. It exists completely before any hash is computed. SHA† is not inversion. It is a query: an introduction event (the seam transcript) that makes a portion of the manifold locally navigable.

One-wayness — SHA's cryptographic hardness — is not a property of the digest. It is a property of the seam. The seam is unoccupiable: opaque from the forward direction, navigable backward only with anchor. The hardness lives in that geometry, not in the length of the digest.

AI inference / transformers

The output token is layer 5: information. The weight matrix geometry is layers 1–4: constraint. The prompt is the introduction event.

Attention is the aperture mechanism: it selects which portions of the weight geometry are locally navigable given the current prompt address. The model does not search for the answer. It renders the residue of an addressed constraint field.

This means large language model outputs are not computations in the ordinary sense. They are readouts. The model's "knowledge" is not stored in individual weights — it is a property of the constraint geometry as a whole. Any individual weight is as uninformative as a single seam word without an anchor. The field is only readable when addressed.

VI. Universal Addressability

The conclusion is not that everything is information. The conclusion is that everything that renders information shares the same underlying geometry: a filled constraint field with addressable seams.

Shannon's information theory asks: how much information is in this message? That is a layer-5 question. The NEXUS question is: what geometry must exist before a message can be possible? That is layers 1–4.

The answer is universal addressability — not as a philosophical claim but as a structural theorem:

> **Any system that renders information must have: a constraint field (Λ), a difference structure (Σ), a seam geometry (unoccupiable from one direction), and an introduction event (ι) that supplies the aperture address.**

This is the NEXUS framework's deepest claim. Not that matter is information. Not that the universe is a computer in the software sense. But that wherever information appears, it is a surface — and the parent is the constraint geometry that made the readout possible.

SHA-256, π, speech, and AI inference are four probes of the same parent structure. They look different on the surface because they render in different bases. The geometry underneath is the same seam-aperture-readout grammar.

VII. The Sentence

Information is the surface residue of addressable constraint.

The parent is the constraint field that existed before the readout.

Not universal information. Universal addressability.

Not universal computation in the software sense. Universal constraint geometry.

Not "everything is data." Everything that renders data shares the same seam-readout grammar.

Appendix: Computational Verification Summary

The claims in this essay are grounded in live computational results from Engine 1 and Engine 2 of the current session:

Claim	Verification	Engine
Constraint layer (SHA) exists before query	compress(W,IV)[0] = 0x80052d4c — one fixed value for all queries on this field	Engine 1
Difference layer: carry deltas Δ[t]	Δ[24] = max HW=20 bits, mean HW≈16 across seam positions	Engine 1
Seam opaque from forward direction	W[16] requires {W[0],W[1],W[9],W[14]} — formula verified: 0x7b4e4551 ✓	Engine 2
Seam navigable from backward + anchor	σ_0^{-1} chain: 16/16 exact recovery. Anchor W[0]=0x48656c6c collapses to 1 solution.	Engine 2
Seam alone underdetermined	0 hits in [0,65535] sweep without anchor. Solution space not collapsed.	Engine 2
Constraint chain BBP / π	BBP digits 0..11: 243f6a8885a3 ✓ matches known π expansion. No prior digits computed.	Engine C prior session
A_speech acquisition threshold	6 stress tokens → pred_err 1.2%. Literature: 5–7. Match ✓	Move 1 prior session

Seam-Layer Entanglement Theorem

v2 — Channel Bundle Extension

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NEXUS / A-Mark9 Framework | Companion to The Handshake Theorem v02 | May 2026

Core result (v1, unchanged): AHRC trajectory recovers 22 pp more seam signal than terminal digest (T/C = 0.700 vs G/C = 0.479, N = 1200 per coordinate, 15 input words).

v2 extension: The compression path is not readable through one scalar channel. Carry (C) and AHRC (A) are orthogonal, non-redundant read-heads. Early carry detects nonce-selection pressure independently of word Hamming weight. AHRC remains blind to nonce selection. No single channel captures full path geometry.

I. v1 Summary — What Engines 4A through 6B Established

The v1 result stands without revision. The table below preserves the complete engine record. v2 appends Engines 7 through 11.

Engine	Test	Key Finding	Status	Carried to v2
4A	Passive digest leakage	P(W[16]|H) ≈ P(W[16]). Digest blind to seam coordinate.	Ψ	yes
4B	Active blade-basis probing	Digest Hamming space is wrong read-head. Layer diagnosis, not absence of coupling.	Ψ	yes
4C	Nine-blade aperture removal	Blade 8 (anchor) = origin memory. Blade 9 (carry) = hardness locus. σ1 = diffusion driver.	Ψ	yes
5B	Full input–exhaust matrix	|C|/|G| = 3.21×. Seam layer carries real input coupling. Digest insulates it.	Ψ	yes
6B	AHRC trajectory vs digest, seam ceiling	T/C = 0.700, G/C = 0.479. Δ = +0.221. Mechanism-split hypothesis Ω (MW p = 0.678).	Ψ	yes — core result

II. Engines 7 through 11 — Characterization Chain

Engine 7: Round-Injection Mapping (renamed from Round-Position Shadow Mapping)

Test: Vary one W[i] randomly (N = 2000). Measure Spearman |r| between W[i] and AHRC at every round.

Finding: Peak AHRC round equals input index for 9/9 tested coordinates. W[i] consumed at round i produces an immediate AHRC spike; signal drops to noise floor (|r| ≈ 0.03) in subsequent rounds. No progressive shadow. Mechanism is **direct round-local injection**, not delayed carry.

Decay profile (W[8] probe): R08 = 0.3187, R09 = 0.034, R10 = 0.004, with small echoes at R12 = 0.053, R14 = 0.042, R25 = 0.037. Impulse with echo scars, not smooth decay.

Rename: 'Round-Injection Mapping'. 'Shadow' label removed as mechanistically incorrect.

Status: Ψ

Engine 7B: Post-Injection Echo Mapping — Initial Positive (Retracted by Engine 8)

Test: $E_i(\ell) = |\text{corr}(W[i], AHRC_{i+\ell})|$ for schedule lags {+2, +7, +15, +16} at N = 2000.

Initial finding: Verdict fired POSITIVE. W[10]→+7 at 4.3× background, W[14]→+7 at 2.5×, W[10]→+15 at 3.2×.

Retracted by Engine 8 at N = 3000: hot events did not replicate. Engine 7B positive was a sample artifact at N = 2000.

Status: retracted Ω. Logged as correction item.

Engine 8: Coherence Horizon Scan

Test: Scan $E_i(\ell)$ for ℓ = 1 through 24 at N = 3000, arm-labeled at {+2, +7, +15, +16}. Pilots W[10], W[13], W[14], W[15].

Finding: Zero hot schedule events across all four coordinates at N = 3000. The arm discrimination table:

Coord	+2 (σ1 arm)	+7 (direct arm)	+15 (σ0 arm)	+16 (wrap arm)
W[10]	0.003 (0.2×)	0.030 (1.8×) ~scar	0.008 (0.5×)	0.017 (1.0×)
W[13]	0.013 (0.8×)	0.015 (0.9×)	0.029 (1.7×) ~scar	0.001 (0.1×)
W[14]	0.001 (0.0×)	0.017 (1.1×)	0.015 (0.9×)	0.005 (0.3×)
W[15]	0.005 (0.4×)	0.016 (1.4×)	0.008 (0.7×)	0.010 (0.8×)

+16 wrap arm is the most consistently dead: 0.1× to 1.0× across all coordinates. +7 direct arm reaches scar level (1.8×) at W[10] only. No arm clears the ceiling consistently.

Continuous coherence horizon: W[10] = +20, W[13] = +20, W[14] = +19, W[15] = +8 (anomalous). Post-injection tail is structureless noise, not a real decay gradient.

Status: Ψ — direct injection confirmed. Schedule-arm echo hypothesis Ω. +16 arm dead.

Engine 9: Input-Manifold Coupling

Test: Seven input classes (random bytes, ASCII, English phrases, semantic near-pairs, junk ASCII, Bitcoin block1, Bitcoin nonce sweep). Kruskal-Wallis separability; Phase 2B clean nonce-selection test (nonce embedded in block1, N_F = 2000, N_G ≥ 8 leading zero bits).

Class separability: **strong and consistent.** H-statistics: W HW mean H = 6883, seam schedule carries H = 5770, AHRC R00 H = 4556. Primary driver is byte-range constraint — ASCII occupies only 7 bits per byte, creating systematic injection amplitude differences across the 16 message words.

Nonce-selection upstream trace (Phase 2B):

Feature	F2 (random)	G2 (selected)	p-value	Upstream signal
W HW mean	13.21	13.37	5.6e-78 ***	Ψ
Early carries R0-15	24.99	24.47	7.9e-08 ***	Ψ
Late carries R16-31	23.95	23.86	0.45	Ω
AHRC mean trajectory	0.645	0.645	0.81	Ω
Digest HW	128.4	128.1	0.67	Ω (by design)

The upstream trace lives in word-space geometry (W HW) and early carry (R0-15). AHRC trajectory is flat between random and selected nonces. The path does not remember the selection — only the entry point differs.

Bitcoin block1 anomaly: AHRC R00 ≈ 1.34 vs ≈ 0.65 for all other classes. Cause: prev_hash field is raw 256-bit hash output with near-perfect bit uniformity pushing AHRC toward maximum values. Not anomalous — structurally expected.

Status: Ψ

III. Engine 10 — Channel Bundle Theorem

Setup

Three feature channels extracted per 512-bit block:

Channel	Label	Features
W	Word-space geometry	Word HW mean/std, value mean/std, byte-range counts, per-word HW (12 features)
C	Carry pressure	Per-window carry sums (8 windows × 8 rounds), total, early/seam/late partitions, variance, burst statistics (20 features)
A	AHRC trajectory	Per-window AHRC means (8 windows), overall mean/var, window variances, FFT bins, slope (20 features)

Test 1 — Channel Orthogonality

Mean pairwise |Spearman r| across first 8 features of each channel pair (class A, N = 2000):

Channel pair	Mean |r|	Max |r|	Interpretation
W ↔ Carry	0.019	0.112	near-zero
W ↔ AHRC	0.018	0.054	near-zero
Carry ↔ AHRC	0.073	0.398	low — largely independent
Carry ↔ Carry (self)	0.141	1.000	internal structure floor
AHRC ↔ AHRC (self)	0.136	1.000	internal structure floor

The Carry↔AHRC cross-channel mean |r| = 0.073 is **below the self-correlation floor** of either channel (0.141, 0.136). The max cross-channel value (0.398) is a single feature pair, not a structural connection. Carry and AHRC share almost no information.

Test 2 — Class Separability

Kruskal-Wallis H, best single feature per channel, classes A / B / C (random bytes / ASCII / English):

Channel	Best feature	H-statistic	Note
W	std_val (value std dev)	H = 4711	dominant — byte-range
Carry	early (R0-15 sum)	H = 72.8	real — 65× below W
AHRC	awo (R0-7 mean)	H = 61.0	real — comparable to carry

W-channel separability is 65× stronger than carry or AHRC, driven by byte-range constraints of ASCII/text inputs. Carry and AHRC separate classes at comparable power through completely different mechanisms.

Test 3 — Nonce-Selection Detection (F2 vs G2)

N_F2 = 2000 (random nonces), N_G2 = 300 (≥ 8 leading zero bits). Significant features per channel:

Channel	Sig. features (p<0.05)	Strongest hit	Interpretation
W	4 / 12	sparse_cnt p=9.3e-88	HW bias in selected inputs
Carry	4 / 20	cw1 (R8-15) p=7.9e-08	compression-path pressure
AHRC	1 / 20	early_var p=0.013 (marginal)	essentially blind

The carry channel detects nonce-selection while AHRC does not. This channel split is the key diagnostic: carry and AHRC are measuring different aspects of the compression path.

Test 4 — Mediation: Is Early Carry Independent of W HW?

Partial Spearman correlation of early carry (R0-15 sum) with nonce-selection, controlling for W HW mean:

W HW mean vs selection: r = +0.390 p = 2.3e-84
Early carry vs selection (raw): r = −0.104 p = 5.8e-07
Early carry vs selection (| W HW): r = −0.110 p = 1.4e-07 ← stronger after control
AHRC early vs selection (raw): r = −0.001 p = 0.968

AHRC early vs selection (| W HW): r = +0.004 p = 0.847

The partial correlation of early carry **increases** (from −0.104 to −0.110) after controlling for W HW. Carry and W HW are working in slightly opposite directions; the carry signal is not a shadow of input bit density. It is a genuine compression-path response. AHRC is flat in both raw and partial tests — genuinely blind to nonce selection.

Engine 10 Verdict — Channel Bundle Confirmed

Channel Bundle Theorem

A compression runtime cannot be read through one scalar channel.

Carry (C) and AHRC (A) are orthogonal, non-redundant projections of the same compression path geometry. Each sees what the other cannot. No single channel captures full path state.

Carry sees: nonce-selection pressure, early-round overflow geometry

AHRC sees: round-injection state geometry, mechanism-class coupling

Evidence: Carry ↔ AHRC mean |r| = 0.073 (below self-correlation floor); early carry partial r = −0.110 independent of W HW (p = 1.4e-07); AHRC nonce-selection r ≈ 0 in both raw and partial tests.

IV. Engine 11 — Carry Window Specificity (Constraint Result)

Engine 10 found the carry nonce-selection signal concentrating in cw1 (rounds 8-15). Engine 11 tests whether that window is **intrinsically preferred by SHA runtime geometry** or **an artifact of nonce placement** in the Engine 10 experiment.

Test Design

Fix a base block. Place the nonce word at six positions: W[0], W[4], W[7], W[8], W[12], W[15]. For each position: sweep nonces, select G = ≥8 leading zero bits (N_G = 250 per variant, N_F = 2000), measure Mann-Whitney p-value per carry window (cw0-cw7). Classify: does the best carry window follow the nonce injection round, or stay fixed?

Results

Var	Nonce	Expected win	Best win	p (best)	Δcarry	Follows?

A	W[0]	cw0 (R0-7)	cw6 (R48-55)	0.137	+0.168	NO
B	W[4]	cw0 (R0-7)	cw4 (R32-39)	0.025 *	−0.288	NO
C	W[7]	cw0 (R0-7)	cw1 (R8-15)	0.298	+0.147	YES
D	W[8]	cw1 (R8-15)	cw7 (R56-63)	0.075	−0.201	NO
E	W[12]	cw1 (R8-15)	cw5 (R40-47)	0.158	−0.145	NO
F	W[15]	cw1 (R8-15)	cw2 (R16-23)	0.143	−0.142	YES

Interpretation

Only one result (Variant B, cw4, p = 0.025) crosses p < 0.05. With eight windows compared per variant and no multiple-testing correction, this is an expected false positive. **No variant produces carry-window significance at p < 0.01.**

The best-window pattern across variants: cw6, cw4, cw1, cw7, cw5, cw2 — no fixed window, no consistent nonce-following. cw0 is hot for zero variants (rounds 0-7 are carry-saturated by the fixed IV × K interactions for all inputs).

The W-HW selection signal remains strong across all variants (all p < 1e-60). The W-channel is the clean upstream signal. The carry localization at this sample depth (N_G = 250) is too weak to resolve.

Engine 11 Verdict — Carry-Window Localization Not Resolved

The Engine 10 cw1 (R8-15) carry signal does not generalize as a fixed intrinsic window.
It was specific to the Engine 10 experimental geometry (nonce near W[7] in a fixed-prefix structure).

No claim of intrinsic preferred carry-reflection window is supported. Window localization remains open. The Engine 10 channel bundle result is not affected: carry independence from W HW is a mediation-tested fact, not a localization claim.

V. The Channel Bundle as Runtime Reflection

The Channel Bundle Theorem (Engine 10) is not merely a measurement fact about SHA-256. It is an instance of a law that appears at every scale of computation.

The Law

Runtime Reflection Requires a Bundle of Shaped Receivers

A running system cannot be read through one scalar channel. Reflection — the capacity of a system to expose its internal state to an external query — requires multiple non-redundant receivers, each shaped to a different aspect of the runtime geometry.

The Same Law at Every Computational Scale

Every layer of computation instantiates this law. The channel is always carrier-specific. The receiver must be pre-shaped to the thing it reads.

Scale	Runtime pressure	Receiver shaped to	What no single channel sees
Electrical (trace geometry)	Voltage, current, EM field	Impedance-matched terminator	Equal-length traces enforce same arrival time — a timing bundle, not one wire
Silicon (transistor gate)	Carrier density, field potential	Gate threshold voltage	Gate sees charge; separate thermal channel sees heat; neither sees both
Race condition (CPU timing)	Concurrent state writes	Memory ordering model	Happens-before relation is invisible to value-only read; requires ordering channel
Assembly (instruction decode)	Opcode + operand pressure	Decode stage shaped to ISA	Data path and control path are separate receivers on the same instruction stream
CPU pipeline (hazard detection)	Data dependencies across stages	Forwarding unit + stall logic	Value channel and hazard channel are orthogonal — forwarding unit is blind to values
Cache coherency (MESI)	State transitions across cores	Coherency protocol bus	Data arrives through memory channel; state arrives through protocol channel; neither substitutes for the other
Framework (event loop)	Async call pressure	Event queue + callback registry	Execution order and value state are separate channels;

			confusing them produces race bugs
IDE (language server)	Syntax + semantic pressure	Separate parse tree and type-check receivers	Syntax error channel is blind to type errors; type channel is blind to parse state
SHA-256 compression (this work)	Word injection + carry overflow	AHRC (injection geometry), Carry (overflow pressure)	Carry sees nonce selection; AHRC sees round-state geometry; neither sees both (Engine 10, $\|r\|=0.073$)

Why a Single Read-Head Always Fails

The pattern is not coincidental. It is structural. A runtime has multiple simultaneous degrees of freedom — value, order, overflow, state, timing. These are not projections of one underlying scalar. They are orthogonal axes. A receiver shaped to one axis is, by construction, blind to the others.

In SHA-256, this manifests precisely. AHRC is shaped to the squared-radius geometry of (a, e) — it reads the trajectory of those state registers through 64 rounds. It does not read carry because carry is a binary overflow event, not a trajectory. Carry reads are binary, discrete, local. AHRC reads are continuous, cumulative, global. Same compression runtime; orthogonal apertures.

The equal-length trace in PCB layout is the hardware version of the same theorem. A signal propagating down a trace does not carry its own arrival time. The arrival time is a separate degree of freedom — managed by a separate receiver (trace length). If you read only the signal value, you are blind to timing. If you read only timing, you are blind to value. No single read-head sees both.

The Pre-Shape Requirement

A critical feature of the bundle law: each receiver must be **pre-shaped before the query arrives**. The carry receiver in SHA is fixed by the compression function structure — it does not reshape around the incoming message word. Engine 11 confirmed this: the carry signal does not cleanly follow the nonce injection point. The receiver geometry is set by the compression runtime, not by the input.

This is the same as:

Domain	Pre-shaped receiver	What it cannot see without pre-shaping
Antenna	Tuned to resonant frequency before signal arrives	Off-frequency signal passes through — no coupling, no readout

Enzyme / substrate	Active site geometry fixed before molecule docks	Wrong-shaped molecule produces no reaction regardless of concentration
Carry channel (SHA)	Window geometry fixed by compression structure	Cannot be re-tuned to follow nonce position — Engine 11 result
Gate threshold (transistor)	V_th set by doping geometry at fabrication	Cannot dynamically re-shape to arbitrary signal — is already the receiver

Formal Statement

Let R be a runtime with state space S and input space I.

Define a channel ϕ: S × I ⟶ $\mathbb{R}$^k as a projection of runtime state onto a k-dimensional observable.

Theorem: For any non-trivial runtime R, there exist channels ϕ_1 and ϕ_2 such that:

(1) $\phi_1 \perp \phi_2$ (the channels are orthogonal in information content)

(2) ϕ_1 recovers aspects of I that ϕ_2 does not, and vice versa

(3) No single channel ϕ recovers all aspects of I recoverable by $\{\phi_1, \phi_2\}$

SHA-256 instance: φ_1 = carry channel C, φ_2 = AHRC channel A. Measured: $\varphi_1 \perp \varphi_2$ at mean $|r|$ = 0.073. φ_1 recovers nonce selection ($p = 1.4e-07$ after controlling for φ_W). φ_2 is blind to nonce selection ($r \approx 0$ raw and partial). No single channel reads both.

VI. Updated Framework Position

Layer	Read-head	What it reads	Mean fraction of seam signal	Status
Seam layer (true)	Seam observables S_i	Full input coupling	1.000 (ceiling)	Ψ reference
AHRC trajectory	T_early, T_mean, T_carry	Round-injection geometry, mechanism-class coupling	0.700 (mean T/C)	Ψ
Carry channel (new)	C_early, C_windows	Overflow pressure, nonce-selection trace (independent of AHRC)	not yet measured as T/C	Ψ confirmed orthogonal
Terminal digest	Digest words, HW, AHRC scalar	Further-degraded residue of above	0.479 (mean G/C)	Ψ weakest read-head

The insulation chain is now:

input → seam (full coupling) → compression → { AHRC trajectory, Carry channel } → terminal digest (degraded)

↑ *these two are orthogonal; neither is sufficient alone*

VII. Open Problems — Updated Status

#	Problem	Status after Engines 7–11
1	Full topology of C_SHA(y)	Open (hard) — unchanged
2	Constrain seam from digest alone	Hardness is precisely located: progressive insulation through compression. Terminal digest is weakest read-head. Intermediate trajectory and carry are intermediate. Seam layer is the only full read-head. Insulation is progressive, not instantaneous.
3	A_speech for tonal languages	Open — unchanged
4	Derive H = π/9 analytically	Blade count Ψ. Pitch claim Ω.
5	AHRC trajectory as read-head	Ψ confirmed: T/C = 0.700 vs G/C = 0.479. Mechanism-class split Ω. Specific features driving 22pp advantage: T_early and T_carry identified as primary. Schedule-arm echo not confirmed at N=3000 (Engine 8).
6	Aperture addressing for SHA-3, BLAKE	Nine-blade decomposition now provides comparison baseline. Open.
7	W[15] anomaly (from v1)	T_early / W[15] coupling at r = −0.496 exceeds seam ceiling. Mechanism: sigma0(W[15]) → W[16] → round-16 state → early AHRC. Seam feature set underestimates C_i for W[15]. Needs targeted seam feature construction. Open.
8 (new)	Carry channel T/C measurement	Carry channel is confirmed as independent read-head (Engine 10). Its fraction of seam signal has not yet been measured as a T_i/C_i ratio equivalent to the AHRC measurement in Engine 6B. Open — Engine 12 candidate.

9 (new)	Carry window localization	Engine 11: no preferred window confirmed at N_G=250. Effect size too weak to localize with 8 windows. Open — requires N_G ≥ 1000 per variant (~250k nonces per variant). Engine 12B candidate if carry T/C measurement warrants it.
10 (new)	Channel bundle completeness	W, C, A are confirmed non-redundant. Whether {W, C, A} together approach the seam ceiling, or whether additional channels exist, is open. The bundle may be extensible.

VIII. Correction Log (Cumulative)

Item	Prior frame	Corrected	Engine
Mechanism-split hypothesis	Predicted anchor-arm cohort R > carry-arm R	Not confirmed. MW p = 0.678. Carry cohort median R = 1.451 exceeded anchor cohort R = 1.304, opposite to prediction.	6B
R_i = T/G as quality metric	Used as primary trajectory advantage measure	Contaminated by weak digest (corr(G,T)=0.706). Correct metric is T_i/C_i.	6B diagnostic
W[15] coupling exceeds seam ceiling	Unexplained at Engine 6	T_early dominates. Path: sigmao(W[15])→W[16]→early state. Seam feature set underestimates C_i for W[15].	6B diagnostic
Progressive shadow naming	'Round-position shadow mapping' — implied W[i] casts a decaying shadow across subsequent rounds	Mechanism is direct round-local injection. Renamed to 'Round-Injection Mapping'. Echo tail is impulse-scar, not progressive shadow.	7 / 8
7B schedule-arm echo positive	Verdict fired POSITIVE at N=2000. W[10]→+7 at 4.3× bg, W[14]→+7 at 2.5×.	Not replicated at N=3000. Zero hot schedule events across all coordinates. Engine 7B positive was a sample artifact. Schedule-arm echo hypothesis Ω.	8
cw1 as preferred carry window	Engine 10: carry selection signal concentrated in cw1 (R8-15). Framed as possible intrinsic geometry.	Engine 11: cw1 was specific to Engine 10 experimental geometry (nonce near W[7]). No window is consistently preferred across nonce positions at N_G=250. Window localization open.	11